

# CM0246 Discrete Structures

## Recursive Definitions and Structural Induction

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# Preliminaries

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## Textbook

Kenneth H. Rosen ([1988] 2004). Matemática Discreta y sus Aplicaciones.

## Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

# Introduction

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## Subjects

- Recursively defined functions from the natural numbers

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- Recursively defined functions from the natural numbers
- Inductively defined sets and structures

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- Inductively defined sets and structures
- Recursively defined functions from inductively defined sets and structures
- Structural induction

# Introduction

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## Subjects

- Recursively defined functions from the natural numbers
- Inductively defined sets and structures
- Recursively defined functions from inductively defined sets and structures
- Structural induction

## Terminological note

We talk about ‘**recursively**’ defined functions and ‘**inductively**’ defined sets/structures. Both terms are used interchangeably in the textbook.

# Introduction

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## Recursion

‘Sometimes it is difficult to define an object explicitly. However, it may be easy to define this object in terms of itself. This process is called **recursion**.’ (Rosen [1988] 2012, 7th ed. p. 344)

# Recursively Defined Functions from the Natural Numbers

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## Recursively defined functions from the natural numbers

Let  $A$  be a set. A function  $f : \mathbb{N} \rightarrow A$  is recursively defined if:

- **Basis step:** We define the function on  $f(0)$ .

# Recursively Defined Functions from the Natural Numbers

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## Recursively defined functions from the natural numbers

Let  $A$  be a set. A function  $f : \mathbb{N} \rightarrow A$  is recursively defined if:

- **Basis step:** We define the function on  $f(0)$ .
- **Recursive step:** We give a rule for  $f(n)$  from the value of  $f$  on smaller natural numbers.

# Recursively Defined Functions from the Natural Numbers

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## Example

$$\begin{aligned}f &: \mathbb{N} \rightarrow \mathbb{N} \\f(0) &= 3, \\f(n+1) &= 2f(n) + 3.\end{aligned}$$

# Recursively Defined Functions from the Natural Numbers

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## Example

$$\begin{aligned}f &: \mathbb{N} \rightarrow \mathbb{N} \\f(0) &= 3, \\f(n+1) &= 2f(n) + 3.\end{aligned}$$

## Example (the factorial function)

$$\begin{aligned}! &: \mathbb{N} \rightarrow \mathbb{N} \\0! &= 1, \\(n+1)! &= (n+1)n!.\end{aligned}$$

# Recursively Defined Functions from the Natural Numbers

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## Example

Give a recursive definition of  $\sum_{k=0}^n a_k$ .

# Recursively Defined Functions from the Natural Numbers

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## Example

Give a recursive definition of  $\sum_{k=0}^n a_k$ .

## Solution

$$\begin{aligned}\sum_{k=0}^0 a_k &= a_0, \\ \sum_{k=0}^{n+1} a_k &= \left( \sum_{k=0}^n a_k \right) + a_{n+1}.\end{aligned}$$

# Recursively Defined Functions from the Natural Numbers

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## Definition

Let  $A$  be a set. A function  $f : \mathbb{N} \rightarrow A$  is **well defined** if for every natural number, the value of the function at this number is determined in an unambiguous way.

# Recursively Defined Functions from the Natural Numbers

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## Problem 56 (p. 254)

Use mathematical induction to prove that a function  $f$  defined by specifying  $f(0)$  and a rule for obtaining  $f(n+1)$  from  $f(n)$  is well defined.

# Recursively Defined Functions from the Natural Numbers

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## Problem 56 (p. 254)

Use mathematical induction to prove that a function  $f$  defined by specifying  $f(0)$  and a rule for obtaining  $f(n+1)$  from  $f(n)$  is well defined.

## Scheme

The function  $f : \mathbb{N} \rightarrow A$  is defined by

$$\begin{aligned}f(0) &= c \in A, \\f(n+1) &= e[f(n)].\end{aligned}$$

# Recursively Defined Functions from the Natural Numbers

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## Problem 56 (p. 254)

Use mathematical induction to prove that a function  $f$  defined by specifying  $f(0)$  and a rule for obtaining  $f(n+1)$  from  $f(n)$  is well defined.

### Scheme

The function  $f : \mathbb{N} \rightarrow A$  is defined by

$$\begin{aligned}f(0) &= c \in A, \\f(n+1) &= e[f(n)].\end{aligned}$$

### Proof

Proof by mathematical induction (see whiteboard).

# Recursively Defined Functions from the Natural Numbers

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## Example (Fibonacci numbers)

$$f : \mathbb{N} \rightarrow \mathbb{N}$$

$$f(0) = 0,$$

$$f(1) = 1,$$

$$f(n+2) = f(n+1) + f(n).$$

# Subjects

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- Recursively defined functions from the natural numbers
- Inductively defined sets and structures
- Recursively defined functions from inductively defined sets and structures
- Structural induction

# Inductively Defined Sets and Structures

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Example (the natural numbers)

See whiteboard.

# Inductively Defined Sets and Structures

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## Example (the natural numbers)

See whiteboard.

## Definition

An **alphabet** is a finite, non-empty set of symbols.

## Example

$$\Sigma_1 = \{0, 1\},$$

$$\Sigma_2 = \{a, b, \dots, z\},$$

$$\Sigma_3 = \{x \mid x \text{ is an Unicode codepoint}\}.$$

# Inductively Defined Sets and Structures

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## Definition

A **string** is a finite sequence of symbols of an alphabet. The empty string is denoted  $\lambda$ .

## Example

See whiteboard.

# Inductively Defined Sets and Structures

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## Definition

A **string** is a finite sequence of symbols of an alphabet. The empty string is denoted  $\lambda$ .

## Example

See whiteboard.

## Example (inductive definition of strings)

See whiteboard.

# Inductively Defined Sets and Structures

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## Definition

A **string** is a finite sequence of symbols of an alphabet. The empty string is denoted  $\lambda$ .

## Example

See whiteboard.

## Example (inductive definition of strings)

See whiteboard.

**Remark:** See the correction to the textbook definition in course's website.

# Inductively Defined Sets and Structures

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## Example (well-formed propositional logic formulae)

- Basis step:  $\text{T}$ ,  $\text{F}$  and  $p$  (propositional variable) are well-formed formulae.
- Inductive step: If  $E$  and  $F$  are well-formed formulae, then  $(\neg E)$ , and  $(E * F)$  with  $* \in \{\wedge, \vee, \rightarrow\}$ , are well-formed formulae.

# Inductively Defined Sets and Structures

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## Inductively defined sets and structures

A set/structure is inductively defined if:

- **Basis step:** We define an initial collection of elements in the set/structure.

# Inductively Defined Sets and Structures

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A set/structure is inductively defined if:

- **Basis step:** We define an initial collection of elements in the set/structure.
- **Inductive step:** We give rules for forming new elements in the set/structure from those already known to be in the set/structure.

# Inductively Defined Sets and Structures

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## Inductively defined sets and structures

A set/structure is inductively defined if:

- **Basis step:** We define an initial collection of elements in the set/structure.
- **Inductive step:** We give rules for forming new elements in the set/structure from those already known to be in the set/structure.
- **Exclusion rule:** We specify that all the elements in the set/structure are those elements specified in the basis step or generated by applications of the inductive step.

# Inductively Defined Sets and Structures

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A set/structure is inductively defined if:

- **Basis step:** We define an initial collection of elements in the set/structure.
- **Inductive step:** We give rules for forming new elements in the set/structure from those already known to be in the set/structure.
- **Exclusion rule:** We specify that all the elements in the set/structure are those elements specified in the basis step or generated by applications of the inductive step.

## Remark

The exclusion rule is tacitly assumed in the slides.

# Subjects

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- Recursively defined functions from the natural numbers
- Inductively defined sets and structures
- Recursively defined functions from inductively defined sets and structures
- Structural induction

# Recursively Defined Functions from Inductively Defined Sets and Structures

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Example (concatenation of strings)

See whiteboard.

# Recursively Defined Functions from Inductively Defined Sets and Structures

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Example (concatenation of strings)

See whiteboard.

Example (length of strings)

See whiteboard.

# Recursively Defined Functions from Inductively Defined Sets and Structures

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## Recursively defined functions from inductively defined sets and structures

A function from an inductively defined set/structure is recursively defined if:

- **Basis step:** We define the function on the initial collection of elements in the set/structure.

# Recursively Defined Functions from Inductively Defined Sets and Structures

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## Recursively defined functions from inductively defined sets and structures

A function from an inductively defined set/structure is recursively defined if:

- **Basis step:** We define the function on the initial collection of elements in the set/structure.
- **Recursive step:** We give rules for defining the value of the function on a new element from those values of the function on the elements of the new element.

# Recursively Defined Functions from Inductively Defined Sets and Structures

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## Recursively defined functions from inductively defined sets and structures

A function from an inductively defined set/structure is recursively defined if:

- **Basis step:** We define the function on the initial collection of elements in the set/structure.
- **Recursive step:** We give rules for defining the value of the function on a new element from those values of the function on the elements of the new element.

### Remark

Note that the exclusion rule is not used in the definition of recursive functions.

# Subjects

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# Structural Induction

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Structural induction is used for proving properties on inductively defined sets/structures.

# Structural Induction

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## Example

Let  $\Sigma$  be an alphabet and  $w \in \Sigma^*$  be a string. Prove that  $\lambda w = w$ .

# Structural Induction

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## Example

Let  $\Sigma$  be an alphabet and  $w \in \Sigma^*$  be a string. Prove that  $\lambda w = w$ .

## Proof by structural induction on $w$

- Basis step: Let  $w = \lambda$ . Then  $\lambda\lambda = \lambda$  by the basis step of  $(\cdot)$ .

# Structural Induction

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## Example

Let  $\Sigma$  be an alphabet and  $w \in \Sigma^*$  be a string. Prove that  $\lambda w = w$ .

## Proof by structural induction on $w$

- Basis step: Let  $w = \lambda$ . Then  $\lambda\lambda = \lambda$  by the basis step of  $(\cdot)$ .
- Inductive step: Let  $w = w'x$  where  $w' \in \Sigma^*$  and  $x \in \Sigma$ .

Inductive hypothesis (IH):  $\lambda w' = w'$ .

$$\begin{aligned}\lambda w &= \lambda \cdot w'x && \text{(by def. of } w\text{)} \\ &= (\lambda \cdot w')x && \text{(by the inductive step of } (\cdot)\text{)} \\ &= w'x && \text{(by IH)} \\ &= w && \text{(by def. of } w\text{)}\end{aligned}$$



# Structural Induction

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## Exercise

Let  $\Sigma$  be an alphabet and  $w, w' \in \Sigma^*$  be two strings. Prove that  $l(ww') = l(w) + l(w')$ .

# Structural Induction

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## Example (left and right parentheses)

Prove that every well-formed propositional formula contains an equal number of left and right parentheses.

# Structural Induction

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## Example (left and right parentheses)

Prove that every well-formed propositional formula contains an equal number of left and right parentheses.

## Proof by structural induction on the set of well-formed propositional formulae

- $P(\alpha)$ :  $l_\alpha = r_\alpha$ , where  $l_\alpha/r_\alpha$  is the number of left/right parentheses in  $\alpha$ .
- Basis step:  $T$ ,  $F$  and  $p$  (propositional variable) have no parentheses, so  $l_\alpha = r_\alpha = 0$ , for  $\alpha \in \{T, F, p\}$ .

(continued on next slide)

# Structural Induction

## Proof (continuation)

- Inductive step

a) Case  $\alpha = (\neg E)$ :

$$\begin{aligned}l_{\alpha} &= 1 + l_E && \text{(def. of } \alpha\text{)} \\&= 1 + r_E && \text{(IH in } E\text{)} \\&= r_{\alpha} && \text{(def. of } \alpha\text{)}\end{aligned}$$

b) Case  $\alpha = (E * F)$ :

$$\begin{aligned}l_{\alpha} &= 1 + l_E + l_F && \text{(def. of } \alpha\text{)} \\&= 1 + r_E + r_F && \text{(IH in } E \text{ and } F\text{)} \\&= r_{\alpha} && \text{(def. of } \alpha\text{)}\end{aligned}$$



# Structural Induction

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## Structural induction

Let  $P$  be a propositional function on an inductively defined set/structure.

To prove that  $P$  is true for all the elements on the set/structure, we must make two proofs:

- **Basis step:** Prove  $P$  for all elements specified in the basis step of the inductive definition of the set/structure.

# Structural Induction

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## Structural induction

Let  $P$  be a propositional function on an inductively defined set/structure.

To prove that  $P$  is true for all the elements on the set/structure, we must make two proofs:

- **Basis step:** Prove  $P$  for all elements specified in the basis step of the inductive definition of the set/structure.
- **Inductive step:** Prove that if  $P$  is true for each of the elements used to construct new elements in the inductive step of the definition, the result holds for these new elements.

# Structural Induction

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## Problem 27(c) (p. 252)

Let  $S$  be the subset of the set of ordered pairs of integers inductively defined by

- Basis step:  $(0, 0) \in S$ .
- Inductive step: If  $(a, b) \in S$ , then
  - a)  $(a, b + 1) \in S$ ,
  - b)  $(a + 1, b + 1) \in S$  and
  - c)  $(a + 2, b + 1) \in S$ .

Use structural induction to show that  $a \leq 2b$  whenever  $(a, b) \in S$ .

(continued on next slide)

# Structural Induction

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Proof by structural induction on  $S$

- $P((a, b))$ : If  $(a, b) \in S$  then  $a \leq 2b$ .

# Structural Induction

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Proof by structural induction on  $S$

- $P((a, b))$ : If  $(a, b) \in S$  then  $a \leq 2b$ .
- Basis step  $P((0, 0))$ :  $0 \leq 2 \cdot 0$

# Structural Induction

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## Proof by structural induction on $S$

- $P((a, b))$ : If  $(a, b) \in S$  then  $a \leq 2b$ .
- Basis step  $P((0, 0))$ :  $0 \leq 2 \cdot 0$
- Inductive step:
  - a)  $P((a, b + 1))$ :

$$\begin{array}{ll} a \leq 2b & (\text{IH } P((a, b))) \\ \leq 2b + 2 & (\text{arithmetic}) \\ \leq 2(b + 1) & (\text{arithmetic}) \end{array}$$

(continued on next slide)

# Structural Induction

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- Inductive step (continuation):

b)  $P((a + 1, b + 1))$ :

$$a \leq 2b$$

(IH  $P((a, b))$ )

$$a + 1 \leq 2b + 2$$

(arithmetic)

$$\leq 2(b + 1)$$

(arithmetic)

# Structural Induction

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- Inductive step (continuation):

b)  $P((a + 1, b + 1))$ :

$$a \leq 2b$$

(IH  $P((a, b))$ )

$$a + 1 \leq 2b + 2$$

(arithmetic)

$$\leq 2(b + 1)$$

(arithmetic)

c)  $P((a + 2, b + 1))$ :

$$a \leq 2b$$

(IH  $P((a, b))$ )

$$a + 2 \leq 2b + 2$$

(arithmetic)

$$\leq 2(b + 1)$$

(arithmetic)



# References

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- Kenneth H. Rosen [1988] (2004). *Matemática Discreta y sus Aplicaciones*. Trans. by José Manuel Pérez Morales, Julio Moro Carreño, Ana Isabel Lías Quintero and Pedro Antonio Ramos Alonc. 5th ed. McGraw-Hill (cit. on p. 2).
- [1988] (2012). *Discrete Mathematics and Its Applications*. 7th ed. McGraw-Hill (cit. on p. 8).